

An Integrated Metaheuristic Approach for Generating Combined Road-Rail Transport Chains

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It is intended that the work is presented in person.

I. INTRODUCTION

Combined transport is a form of intermodal freight transport designed to combine the respective advantages of different transport modes. Specifically, in combined road-rail transport, freight is moved in three legs: the pre-haul from the origin to the departure terminal, the main haul between terminals, and the post-haul from the arrival terminal to the final destination. The road-based pre- and post-haul legs, where loading units (LUs) are transported individually, are kept as short as possible, while the longest part of the journey is performed by rail, where units are consolidated on trains [9].

Along the transport chain, coordination among shippers, freight forwarders, intermodal transport service providers, terminal operators, and rail operators is required. However, each actor typically has access only to information within their own operational scope.

In this study, we use empirical transport data provided by a combined transport service operator. This dataset contains detailed information on LUs transported on the rail main leg, but it does not include the preceding first-mile and subsequent last-mile road movements. As a result, the dataset represents only a segment of the intermodal transport chain and does not capture complete origin-to-destination movements, which limits the analysis of operational interactions between road and rail transport.

Against this background, this study develops a methodological framework to generate complete intermodal freight transport chains. This framework augments historical combined transport data of the rail-based main haulage with synthetic first- and last-mile road segments, derived from OpenStreetMap (OSM) data [4]. The resulting complete chains enable scenario analyses and sensitivity testing using agent-based simulation approaches [8] such as MATSim [2, 3].

II. METHODOLOGY

The methodological framework consists of four sequential steps: (i) chain construction, (ii) plan generation, (iii) plan execution, and (iv) results evaluation, as illustrated in Figure 1. First, the chain construction step augments the rail-based freight transport data with synthetic first- and last-mile road segments to create complete combined transport chains. Second, during plan generation, an optimisation heuristic, Adaptive Large Neighborhood Search (ALNS), determines a

route for each LU. Third, the resulting routes are executed in MATSim during the plan execution step. Finally, the results evaluation step assesses the simulation outputs and supports the comparison of scenarios and sensitivity analyses.

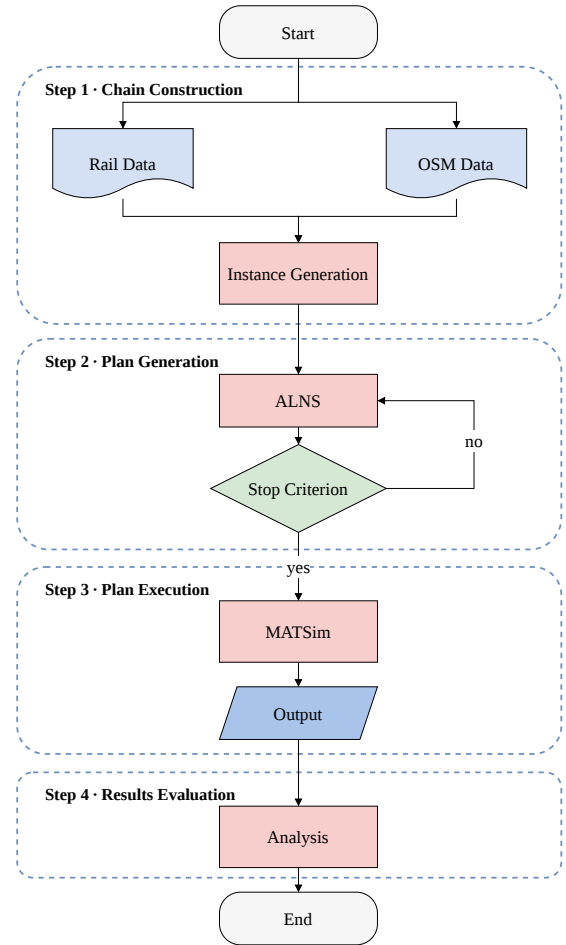


Figure 1. Process diagram of the multi-step framework.

A. Chain Construction

The chain construction step first generates a complete intermodal transport chain for each LU to be transported. Since the empirical dataset contains only the rail-based main

leg, synthetic first- and last-mile road segments are added to connect the observed rail freight transport to an origin and destination. The step then generates a set of plans for each LU. The resulting output is an instance of complete combined transport chains enriched with departure and arrival times for every transport leg.

OSM data are used to identify potential origins and destinations within the catchment areas of the relevant combined transport terminals. For each LU in the rail-only dataset, one origin and one destination are randomly selected from the respective candidate locations to represent the first- and last-mile road freight transports. For the catchment area definition, a radius of 50 km around each combined transport terminal is assumed [11]. The resulting chain, depicted in Figure 2, therefore connects a synthetic origin and destination to the observed origin and destination terminals.

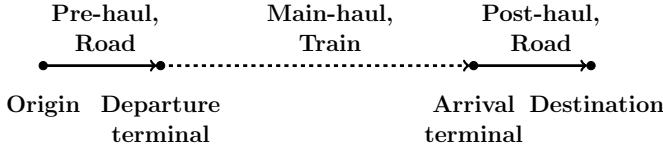


Figure 2. Combined intermodal road-rail transport chain.

To construct the candidate-location set, OSM features are selected using predefined tag combinations representing logistics-relevant and industrial land uses, including warehouses, distribution centres, and manufacturing facilities. The extraction is performed from the Geofabrik Germany PBF [1] dataset using QuackOSM [7]. In contrast to conventional workflows that may require loading large OSM extracts into memory, QuackOSM supports the spatial filtering of PBF data through DuckDB and its spatial extension. This approach facilitates the processing of the large study area and the extraction of facilities located within the terminal catchment areas.

The resulting locations constitute synthetic representations of potential first- and last-mile origins and destinations rather than observations of actual shipper or consignee locations. Their validity is consequently conditional on the quality of the underlying OSM data. Spatial coverage, positional accuracy, semantic consistency of tags, and attribute completeness may vary between regions and facility types. Moreover, operationally relevant attributes, such as facility capacity, opening hours, or actual freight flows, are generally not available in OSM. The generated locations should therefore be interpreted as plausible candidate locations for scenario-based analysis rather than as a complete representation of the regional logistics landscape.

Subsequently, a list of possible departure dates serve as the basis for generating plans for each LU. These plans enrich the transport chain with departure and arrival times for each leg and provide alternative inputs for the subsequent plan generation step.

B. Plan Generation

To generate alternative initial plans for each LU, an attribute specifying the possible departure days for each LU and possible departure day at its origin terminal, the rail service schedule is queried for a direct connection to the destination terminal. When a direct service is available, the resulting plan consists of the waiting time between the start date and the departure of the respective service, followed by the direct rail movement. Where no direct service is available, a time-dependent shortest-path algorithm, accounting for transfer times is applied based on Pyrga et. al. [6]. An additional plan is only added, if the resulting shortest path does not consist of waiting until a later day to use a direct service. To ensure compatibility, each plan is assigned a score based on the waiting time at the origin terminal and the travel time of the respective plan.

To determine a plan combination, that dispatches as many LUs as early as possible, an ALNS is implemented [10]. In this study, the objective is to select one plan for each LU such that the overall sum of the associated plan scores is minimised. During the ALNS, an initial solution is iteratively destroyed and repaired. When a solution with a lower value than the current objective value is found in an iteration, this solution replaces the existing best solution [5].

The destroy operator randomly removes a number of LUs proportional to a 50% degree of destruction, while the repair operator reinserts the plan with the lowest score from the memory of plans, which does not violate the rail services capacities. ALNS also allows the implementation of additional repair and destroy operators [5], such as removing the LUs with the worst scores or rerouting the LUs on the transit schedule. During initialisation of the ALNS, the LUs select a plan randomly from their memory of alternative plans, that is feasible. These selected plans serve as the initial solution to be optimised. The predefined number of iterations functions as a stop criterion. The selected plan for each LU specifies the earliest departure time at the origin terminal and the latest arrival time at the destination terminal. Together with the first- and last-mile road segments, these values enable the determination of the latest delivery and the earliest collection times at the terminals.

III. PRELIMINARY RESULTS

To generate an experimental instance, a sample of ten LUs is drawn from the observed rail freight transport data. Each sample represents a LU and includes relevant attributes, such as weight, length, origin and destination terminals and preferred departure and arrival times. In the generated instance, each LU is assigned a unique set of attributes. Following the chain construction step, we derive an origin and a destination around its origin and destination terminal within a radius of 50km. Figure 3 illustrates the terminals and the candidate nodes located within their corresponding catchment areas.

With respect to transit schedule, the scope of this study is restricted to rail services between German combined transport terminals. Only services with both an origin and a destination

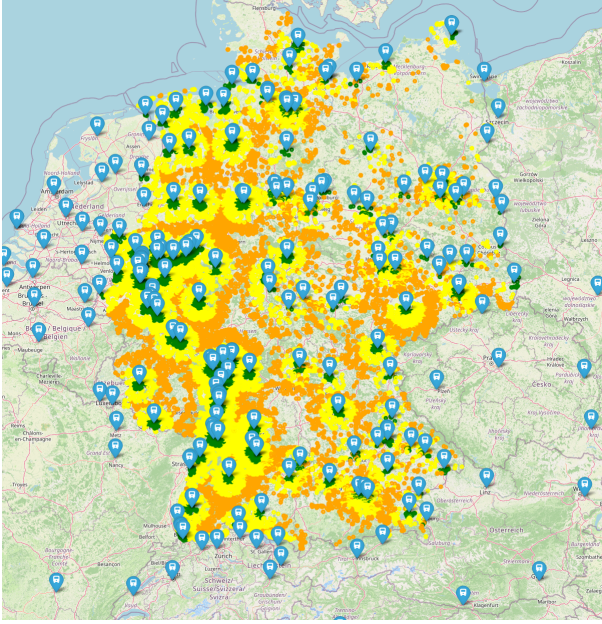


Figure 3. Terminals and candidate first- and last-mile nodes within their 50 km catchment areas. Terminals are shown as blue markers; candidate nodes within 0–10 km, 10–30 km, and 30–50 km are shown in green, yellow, and orange, respectively.

terminal in Germany are considered. The capacity of each rail service is constrained by the maximum permissible length and weight. The ALNS is executed for ten iterations. Given the small sample size in the test instance, it was expected that each LU can select the plan with the earliest departure time. This expectation was confirmed by the metaheuristic optimisation results. The integration of the framework is ongoing and the presentation of results from experimental instances including a MATSim run will be delivered at the MATSim User Meeting.

IV. CONCLUSION

In this study, a methodological framework for constructing complete combined intermodal road-rail transport chains from publicly available and historical rail freight transport data was presented. By integrating synthetic first- and last-mile road segments, with these data, the framework accounts for operational components that are not captured by the rail main leg alone. The methodology is structured as a multi-step procedure comprising chain construction, plan generation, plan execution and evaluation of the results. The resulting framework provides a scalable basis for future scenario analyses and sensitivity tests of intermodal freight operations. In the next steps of this study, the methodological framework will be integrated fully to overcome current fragmentation. This will facilitate a seamless, end-to-end workflow that enables detailed evaluations of intermodal freight systems.

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